

Missing Number

Given a permutation of numbers from 1 to N .

1. A number has been picked and discarded.
2. The remaining numbers are shuffled.
3. Some (possibly zero) of the numbers have been added with negative signs.
4. An increment of constant integer K , where $-3N \leq K \leq 3N$, is added to each number.

Find the **sum** of all possible values of the **missing number**.

If there are different K values that lead to the same missing number, those missing numbers should be counted again in the sum. For example, if there are 2 different K 's that lead to the same missing number, 5, then the answer should be $5 + 5 = 10$ (assuming no other K 's lead to a valid missing number/configuration).

Note:

1. Only $N - 1$ integers are given in the input since the missing number is discarded.
2. A sequence of p integers is called a permutation if it contains all integers from 1 to p exactly once.

Constraints

- $1 \leq N \leq 100000$,
- $-3N \leq K \leq 3N$.

Sample Input 1

```
N = 5
Numbers = [-1, 7, 4, 1]
```

Sample Output 1

Sample Input 2

$N = 6$
 Numbers = [4, 5, 13, 6, 11]

Sample Output 2

7

Explanation:

For Sample 1, from permutation [1, 2, 3, 4, 5], '4' is discarded before shuffling remaining numbers and negative signs are added to '1' and '3'. [-3, 5, 2, -1] becomes [-1, 7, 4, 1] after an increment of 2. The only possible value of the missing number is 4.

For Sample 2, there are two possible answers for the missing number. When K is 10, the missing number is 2 and when K is 7, the missing number is 5. Their sum is 7.

Solution

There are 4 general cases to solve this problem. Let's look at the largest number in the permutation, which is N . If we put a negative sign to N , it must be the smallest number of the final sequence regardless of the addition of K . If we don't add a negative sign to N , it must be the largest number of the final sequence. When the missing number is N , $N - 1$ is the largest number in the permutation, two cases are also considered here whether we place a negative sign to $N - 1$.

Denote a valid sequence a sequence which has $N - 1$ numbers when all the magnitudes are different and the magnitudes are in the range of $[1, N]$.

Example: $N = 6$, sequence = [-3, 5, -2, 1, 6]

Approach:

1. Determine K .
2. Obtain the sequence before addition of K .
3. Check the validity of the sequence.
4. Find the missing number.

Case 1: Missing number is not N , N has a negative sign. Hence, $-N$ is the smallest before and after the addition of K .

Find the minimum number of the final sequence, min , and determine the constant K by applying $-N + K = min$.

Case 2: Missing number is not N , N doesn't have a negative sign. Hence N is the largest before and after the addition of K .

Find the maximum number of the final sequence, max , and determine the constant K by applying $N + K = max$.

Case 3: Missing number is N , $N-1$ has a negative sign. Hence, $-(N-1)$ is the smallest before and after the addition of K .

Find the minimum number of the final sequence, min , and determine the constant K by applying $-(N - 1) + K = min$.

Case 4: Missing number is N , $N-1$ doesn't have a negative sign. Hence $N-1$ is the largest before and after the addition of K .

Find the maximum number of the final sequence, max , and determine the constant K by applying $(N - 1) + K = max$.

After accumulating all possible values of K , we check the validity of sequence before the addition of K . If it is valid sequence, we could find the missing number easily in the sequence (before addition of K). The missing number, x , $x \in [1, N]$ does not appear in the sequence (before addition of K) as either x or $-x$.

Example: $N = 6$, sequence = $[-3, 5, -2, 1, 6]$

4 is the missing number as it the sequence does not contain -4 and 4 .

An efficient way to check the missing number of a valid sequence is taking the sum of from 1 to N and minus the absolute value of each element in the valid sequence.

Finally, we sum all the possible values of the missing number.

Implementation

```
n = 5
numbers = [-1, 7, 4, 1]

numbers.sort()

mn = numbers[0]
mx = numbers[-1]

kk = set()

#Case 1: missing number is not n
k = mx - n
kk.add(k)

#Case 2: missing number is not n, negative sign put on n
k = mn - (-n)
kk.add(k)

#Case 3: missing number is n
k = mx - (n - 1)
kk.add(k)

#Case 4: missing number is n, negative sign put on n-1
k = mn -(-(n - 1))
kk.add(k)

ans = 0

for k in kk:
    #check validity of sequence
    arr = []

    for x in numbers:
        arr.append(x - k)

    s = set() #avoid duplicates
    total = n * (n + 1) / 2

    isValid = True

    for x in arr:
```

```
value = abs(x)
if value < 1 or value > n:
    #out of range
    isValid = False
    break

if value in s:
    #found duplicate
    isValid = False
    break

s.add(value)
total -= value

if isValid:
    ans += total

print(ans)
```